Medians and quartiles

9C Threat from disease · Lesson 5 of 7

Date _____ Class _____

Today I can

- ☐ I can find the median, the lower and upper quartiles and the interquartile range of a simple data set.
- ☐ I can use quartiles to compare the variation in two groups and to interpret the results of a medicine trial.

What happens today

1. Do Now: nine pulse rates
2. Learn: Median and quartiles · Why use the interquartile range? · and more
3. Practical: Class data: hand span
4. Challenge, then the exit ticket

Student Book pages 46 to 47 · Workbook pages 33

What I learned today (one or two sentences)

Key Vocabulary

Term	Definition
median	the middle value when the values are written in order
mean	the total of the values divided by the number of values
range	the largest value minus the smallest value
quartile	a value one quarter or three quarters of the way through the ordered data
lower quartile	the value one quarter of the way through the ordered data (LQ)
upper quartile	the value three quarters of the way through the ordered data (UQ)
interquartile range	UQ minus LQ: the spread of the middle 50% of the data
normal distribution	the bell-shaped curve seen when a continuous variable is plotted for a population
continuous variation	variation that can take any value in a range, e.g. height
clinical trial	a test of a medicine on groups of people

Quick Review

nine pulse rates

Resting pulse rates (beats per minute) of nine students: 72, 65, 80, 58, 90, 66, 75, 61, 70.

1 Write the values in order, smallest first.

2 Which value is in the middle?

3 Calculate the range (largest minus smallest).

Core 1 of 4

Everyone does these. Answer on the card or under it.

1 Nine ladybirds were counted for spots: 7, 2, 9, 4, 7, 5, 2, 8, 6.

a) Write the values in order, smallest first.

b) State the median.

c) Calculate the range.

Core 2 of 4

Everyone does these. Answer on the card or under it.

2 The foot lengths (cm) of 36 male students were: 25.7, 29.1, 26.4, 26.5, 28.2, 27.9, 27.9, 28.1, 26.9, 27.3, 27.2, 28.8, 27.9, 27.4, 28.2, 26.6, 27.3, 27.8, 26.3, 27.3, 28.0, 26.6, 27.1, 26.6, 26.4, 25.9, 27.3, 25.1, 28.5, 26.7, 28.8, 26.6, 28.6, 29.0, 25.7, 28.9.

a) Complete the table by counting the number of students in each group. Tick off each value as you use it.

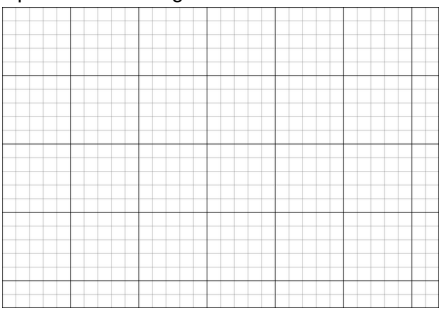
Foot length (cm)	25.1 to 25.5	25.6 to 26.0	26.1 to 26.5	26.6 to 27.0	27.1 to 27.5	27.6 to 28.0	28.1 to 28.5	28.6 to 29.0	29.1 to 29.5
Number of students									

Core 3 of 4

Everyone does these. Answer on the card or under it.

2 *continued*

b) Plot a bar chart of the grouped data on the grid. The bars should touch. Label both axes.



c) Identify the group that contains the median foot length.

Core 4 of 4

Everyone does these. Answer on the card or under it.

2 *continued*

d) Does your bar chart show a normal distribution? Explain your answer.

Challenge 1 of 3

When Core is finished. Full sentences.

C1 In a statin trial, patients were measured for their ability to absorb cholesterol and split into quartiles. The table shows the change in their risk of a heart attack after taking statins.

Quartile of cholesterol absorption	lowest	lower middle	upper middle	top
Change in risk of heart attack (%)	-37.7	-34.3	-24.7	+16.6

a) Explain why the data for cholesterol absorption were split into quartiles.

Challenge 2 of 3

When Core is finished. Full sentences.

C1 *continued*

b) The quartile boundaries of the cholesterol absorption index were 107, 126 and 148. Calculate the interquartile range.

c) Explain what the results show about the effect of cholesterol absorption on how well statins reduce the risk of a heart attack.

Challenge 3 of 3

When Core is finished. Full sentences.

C2 Explain why the interquartile range is more useful than the range for comparing the variation in two different groups.

Exit Ticket

Hand this card in at the end of the lesson.

Name _____

1 Write these values in order and give the median: 12, 5, 9, 20, 7.

2 The upper quartile is 30 and the lower quartile is 18. Calculate the interquartile range.

3 What does a small interquartile range tell you about a population?

Today I can: Goal 1 ☐ Goal 2 ☐